

Crystal planes in cubic cells: how to see and slice them

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1. Basics: coordinates and atoms in the cube

I put the origin at one corner of the cube and measure every position in units of the lattice parameter a . The cube then runs from 0 to 1 along x , y and z , and I write each atom as fractional coordinates (x, y, z) .

I draw these atoms in one cube (every ball a textbook picture shows):

Structure	Atom positions I draw	Balls drawn	Atoms the cell owns
Simple cubic (SC)	8 corners	8	$8 \times 1/8 = 1$
BCC	8 corners + body center $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$	9	$1 + 1 = 2$
FCC	8 corners + 6 face centers	14	$1 + 6 \times \frac{1}{2} = 4$

The FCC face centers sit at $(\frac{1}{2}, \frac{1}{2}, 0)$, $(\frac{1}{2}, \frac{1}{2}, 1)$, $(\frac{1}{2}, 0, \frac{1}{2})$, $(\frac{1}{2}, 1, \frac{1}{2})$, $(0, \frac{1}{2}, \frac{1}{2})$ and $(1, \frac{1}{2}, \frac{1}{2})$.

Drawn versus owned: 8 cubes share each corner atom and 2 cubes share each face atom, so the FCC cell owns 4 atoms. When I draw, slice or project a cube, I always work with all 14 balls. "4 atoms" and "14 balls" are both right; they count different things.

2. Miller indices and the plane equation

I keep two things apart: h, k, l belong to the plane, and x, y, z belong to an atom. The Miller indices (hkl) are three fixed numbers that name a plane family. The coordinates (x, y, z) change from atom to atom.

How I get the equation:

1. The plane (hkl) cuts the x, y and z axes at $1/h, 1/k$ and $1/l$. An index of 0 means the plane never cuts that axis (intercept at infinity).
2. Any flat plane that cuts the axes at p, q and r follows the intercept form $x/p + y/q + z/r = 1$.

3. I substitute $p = 1/h$, $q = 1/k$, $r = 1/l$.

Plane equation: $h \cdot x + k \cdot y + l \cdot z = 1$

My checks:

- (111) cuts all three axes at 1, so I get $x + y + z = 1$. This is the yellow triangle through (1,0,0), (0,1,0) and (0,0,1).
- (001) never cuts x or y and cuts z at 1, so I get $0 \cdot x + 0 \cdot y + 1 \cdot z = 1$, which is $z = 1$: the top face of the cube.
- (110) cuts x and y at 1 and never cuts z, so I get $x + y = 1$: a vertical rectangle through the cube diagonal.

Negative index: I shift the origin to a neighboring corner, or I keep the minus sign in the formula. The sorting method in section 3 works the same either way.

3. Slices: sorting atoms onto parallel planes

Every plane parallel to (hkl) keeps the same left side and changes only the number on the right. I treat that right-hand number as a slice number.

Parallel slices: $h \cdot x + k \cdot y + l \cdot z = s$, where $s = 0, \frac{1}{2}, 1, 1.5, 2$, and so on

How I use it:

1. I take one atom's coordinates (x, y, z).
2. I compute $s = hx + ky + lz$.
3. Atoms with the same s share a slice. Atoms with different s sit on different, parallel slices.

My examples for (111), where $s = x + y + z$:

- (0,0,0) gives 0, so it sits on slice 0.
- ($\frac{1}{2}, \frac{1}{2}, 0$) gives 1, so it sits on slice 1, the yellow triangle.
- (1,1,0) gives 2, so it sits on slice 2.

Special case (001): $s = z$, so the slice number is just the atom's height. I picture the slices as floors of a building.

In this note, "layer" and "slice" mean the same thing: one flat sheet of atoms parallel to the (hkl) plane.

4. How many slices, and how far apart

I never choose the slices; the atoms decide them. A slice exists only where an atom sits.

1. I compute $s = hx + ky + lz$ for every ball in the cube.
2. I count the different values of s . That count gives the number of slices inside the cube.
3. I take the smallest gap Δs between neighboring values.
4. I convert that gap to a distance.

Spacing: $d = \Delta s \times a / \sqrt{h^2 + k^2 + l^2}$

My FCC results:

Plane	Values of s	Slices in the cube	Smallest gap Δs	Spacing d	Balls per slice
(001)	0, $\frac{1}{2}$, 1	3	$\frac{1}{2}$	$a/2 = 0.500a$	5, 4, 5
(110)	0, $\frac{1}{2}$, 1, 1.5, 2	5	$\frac{1}{2}$	$a/(2\sqrt{2}) = 0.354a$	2, 2, 6, 2, 2
(111)	0, 1, 2, 3	4	1	$a/\sqrt{3} = 0.577a$	1, 6, 6, 1

These spacings match the FCC d -spacings I see in XRD: $d(111) = a/\sqrt{3}$, $d(200) = a/2$, $d(220) = a/(2\sqrt{2})$. This explains why FCC shows the (200) and (220) peaks instead of (100) and (110): the real slices sit twice as close as the (100) and (110) labels suggest.

Why I use the smallest gap: one cube does not hold every slice of the crystal. In BCC (111), the cube gives $s = 0, 1, 1.5, 2, 3$, so the gaps come out uneven. Body centers of neighboring cubes fill the missing values (0.5, 2.5). The smallest gap, $\frac{1}{2}$, gives the true spacing $a/(2\sqrt{3}) = 0.289a$.

Edge slices: the slice at the top of one cube repeats the slice at the bottom of the next. For FCC (001), $z = 1$ repeats $z = 0$, so only two kinds of slice alternate: corner-type and middle.

5. Two ways to look at a plane

Every plane figure shows one of two pictures. I got confused before because I did not know which one I was looking at.

	Picture A: single slice	Picture B: projection
What I keep	Only the atoms with one value of s	All atoms in the cell, every slice

	Picture A: single slice	Picture B: projection
How I look	Straight at the sheet	Straight down $[hkl]$, through all slices
Distances I see	True distances	True only within one slice; atoms from different slices look closer than they are
Overlaps	None	Atoms on one line of sight land on one spot
Where I meet it	Planar density, textbook plane drawings, the top layer of a cleaved surface	Journal figure insets, "view along $[hkl]$ " images

Why I look along $[hkl]$: the (111) plane is perpendicular to the $[111]$ direction

I keep two notations apart. Round brackets (111) name a plane. Square brackets $[111]$ name a direction: the arrow from $(0,0,0)$ to $(1,1,1)$, the body diagonal of the cube.

In a cubic cell, the direction $[hkl]$ always stands perpendicular to the plane (hkl) . For (111) , I check it with two lines that lie inside the yellow triangle:

- From $(0,1,0)$ to $(1,0,0)$: the line vector is $(1,-1,0)$. Its dot product with $(1,1,1)$ is $1 - 1 + 0 = 0$.
- From $(0,0,1)$ to $(1,0,0)$: the line vector is $(1,0,-1)$. Its dot product with $(1,1,1)$ is $1 + 0 - 1 = 0$.

A zero dot product means 90° . The arrow $[111]$ meets two different lines of the plane at 90° , so it stands perpendicular to the whole plane.

The general reason: the plane equation $h \cdot x + k \cdot y + l \cdot z = s$ holds for every point on the plane. If I take any two points on it and subtract, I get $h \cdot \Delta x + k \cdot \Delta y + l \cdot \Delta z = 0$. So the vector (h, k, l) has a zero dot product with every line inside the plane. The numbers h, k, l in the plane equation are the components of the plane's normal.

What this gives me:

- To see the (111) slices face-on, I look along $[111]$. Every slice then lies flat in front of me, and the slices stack straight behind one another.
- Walking along $[111]$ from $(0,0,0)$ to $(1,1,1)$, I cross the slices $s = 0, 1, 2$ and 3 one after another. Three gaps of $a/\sqrt{3}$ add up to $a\sqrt{3}$, the length of the body diagonal. This checks the spacing formula.

- The value $s = hx + ky + lz$ grows only when I move along $[hkl]$, and stays fixed when I move inside a slice. That makes s a ruler along the normal, and dividing by $\sqrt{h^2 + k^2 + l^2}$ turns it into a real distance.

Cubic only: the rule works because the cube axes stand at 90° and share one length a . In hexagonal, tetragonal or other non-cubic cells, $[hkl]$ and the normal to (hkl) generally point in different directions.

Which slice I pick for picture A: in a one-atom-per-lattice-point metal (Cu, Al, W), every slice of a family carries the same atom pattern, only shifted sideways. Any slice gives the same picture. By convention I draw the slice with $s = 1$, because it cuts the axes at $1/h$, $1/k$, $1/l$ and fits inside one cube.

The sideways shift between slices is the stacking sequence. In FCC (111), three shifted positions repeat: ABC stacking.

How I spot a projection: if a figure shows atoms that cannot all satisfy the same value of $hx + ky + lz$, I know it shows picture B.

6. Worked examples: FCC (001), (110), (111)

FCC (001): $s = z$

s	Balls	Count
0	(0,0,0), (1,0,0), (0,1,0), (1,1,0), ($\frac{1}{2}, \frac{1}{2}, 0$)	5
$\frac{1}{2}$	($\frac{1}{2}, 0, \frac{1}{2}$), ($\frac{1}{2}, 1, \frac{1}{2}$), ($0, \frac{1}{2}, \frac{1}{2}$), ($1, \frac{1}{2}, \frac{1}{2}$)	4
1	(0,0,1), (1,0,1), (0,1,1), (1,1,1), ($\frac{1}{2}, \frac{1}{2}, 1$)	5

- Picture A (slice $z = 0$): I see an $a \times a$ square with 4 corners and 1 center ball. The nearest-neighbor distance is $a/\sqrt{2}$, corner to center.
- Picture B (looking down z): slices $z = 0$ and $z = 1$ fall exactly on each other. I see 9 spots: 4 corners, 1 center, and 4 edge midpoints from slice $z = \frac{1}{2}$. The $z = \frac{1}{2}$ balls form a square of side $a/\sqrt{2}$, rotated 45° .

FCC (110): $s = x + y$

s	Balls	Count
0	(0,0,0), (0,0,1)	2
$\frac{1}{2}$	($\frac{1}{2}$,0, $\frac{1}{2}$), (0, $\frac{1}{2}$, $\frac{1}{2}$)	2
1	(1,0,0), (0,1,0), (1,0,1), (0,1,1), ($\frac{1}{2}$, $\frac{1}{2}$,0), ($\frac{1}{2}$, $\frac{1}{2}$,1)	6
1.5	(1, $\frac{1}{2}$, $\frac{1}{2}$), ($\frac{1}{2}$,1, $\frac{1}{2}$)	2
2	(1,1,0), (1,1,1)	2

- Picture A (slice $s = 1$): I see a rectangle $a\sqrt{2}$ wide and a tall, with balls at the 4 corners and at the middle of the top and bottom edges. Spacing along the long edge is $a/\sqrt{2}$.
- Picture B (looking down [110]): I see 8 spots in an $a\sqrt{2} \times a$ rectangle. The bottom row has 3 spots, and its middle spot stacks 3 balls: (0,0,0), (1,1,0) and ($\frac{1}{2}$, $\frac{1}{2}$,0). The top row repeats this at $z = 1$. The middle row has 2 spots at $\pm 0.354a$ from center, each holding 2 balls from slices $\frac{1}{2}$ and 1.5.

FCC (111): $s = x + y + z$

s	Balls	Count
0	(0,0,0)	1
1	(1,0,0), (0,1,0), (0,0,1), ($\frac{1}{2}$, $\frac{1}{2}$,0), ($\frac{1}{2}$,0, $\frac{1}{2}$), (0, $\frac{1}{2}$, $\frac{1}{2}$)	6
2	(1,1,0), (1,0,1), (0,1,1), (1, $\frac{1}{2}$, $\frac{1}{2}$), ($\frac{1}{2}$,1, $\frac{1}{2}$), ($\frac{1}{2}$, $\frac{1}{2}$,1)	6
3	(1,1,1)	1

- Picture A (slice $s = 1$): I see an equilateral triangle of side $a\sqrt{2}$, with corner balls at its tips and face-center balls at the midpoints of its edges. The nearest-neighbor distance is $a/\sqrt{2}$.
- Picture B (looking down [111], the body diagonal): the four slices stack into one image.
 - Slice 1 shows as the triangle at true size.
 - Slice 2 shows the same triangle turned 180° . Together they form a six-pointed star.
 - Slices 0 and 3 are single corners on my line of sight, so both land on the center spot.

- Result: a hexagon (the cube's outline seen corner-on) with 6 tip spots at radius $a\sqrt{2/3} = 0.816a$, 6 inner spots at $a/\sqrt{6} = 0.408a$, and 1 center spot holding 2 balls. 13 spots, 14 balls.

The 2D repeat cell of a (111) slice

The triangle is only the part of the slice that the cube walls cut off. It cannot serve as a unit cell, because triangles do not tile a plane by translation alone. The smallest repeat unit is a rhombus with side $a/\sqrt{2}$ and angles of 60° and 120° , holding 1 atom. My example: $(1,0,0)$, $(\frac{1}{2}, \frac{1}{2}, 0)$, $(0, \frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, 0, \frac{1}{2})$, all from slice 1.

My check: the triangle has area $\sqrt{3}a^2/2$ and holds $3 \times 1/6 + 3 \times 1/2 = 2$ atoms. The rhombus has area $\sqrt{3}a^2/4$ and holds 1 atom. One triangle equals two rhombi.

7. Reading a plane figure someone shows me

My checklist:

1. I identify the structure (SC, BCC, FCC) and list the balls of one cube with their coordinates.
2. I compute $s = hx + ky + lz$ for each ball and group the balls into slices.
3. I decide whether the figure shows picture A (one slice) or picture B (projection). Atoms from more than one slice mean picture B.
4. I count spots. For picture B, I compare spots with balls: missing balls hide behind others on the same line of sight.
5. I check distances. Within one slice, the nearest FCC distance is $a/\sqrt{2}$. Any labeled distance or drawn cell should join atoms of the same slice.
6. For a cleaved surface or a 2D cell, I expect a parallelogram (a rhombus for (111)), not the triangle.

Case: Materials Studio cleave

- Cleave (111) builds a periodic slab, so its in-plane cell is the rhombus: $a = b = a(\text{FCC})/\sqrt{2}$, $\gamma = 120^\circ$ (or 60°). It never shows the triangle.
- The window shows periodic images on the cell edges and may show several slices, so the ball count differs from one slice.
- To compare with a textbook projection, I use a single-element FCC conventional cell and view exactly along $[111]$. In VESTA, I set the projection vector $u\ v\ w$ in the Orientation dialog.

- NaCl (rock salt) consists of two interpenetrating FCC lattices, so its pictures differ from single-element FCC.

8. Primitive and conventional cells

A unit cell is any box that rebuilds the whole crystal when I copy it and shift it, without rotating it. Many boxes pass this test. I keep two of them in mind:

	Conventional cell	Primitive cell
Purpose	Shows the symmetry (cube edges, 90° angles)	Smallest box that still works
Lattice points inside	Can be more than 1 (FCC 4, BCC 2)	Always exactly 1
Shape for cubic crystals	Always the cube	SC: the cube. BCC and FCC: a slanted box (rhombohedron)
Where I use it	Miller indices, textbook drawings, XRD indexing	DFT (fewer atoms, cheaper), band structures, Brillouin zones

How I build a 3D primitive cell

1. I pick one atom as the origin.
2. I draw vectors to three of its nearest neighbors that do not lie in one plane.
3. I complete the slanted box (parallelepiped) on those three vectors.
4. I check it: the box holds exactly 1 atom ($8 \text{ corners} \times 1/8$), and its volume equals the conventional volume divided by the atoms in the conventional cell.

Structure	Primitive vectors (units of a)	Vector length	Angle between vectors	Volume
SC	$(1,0,0), (0,1,0), (0,0,1)$	a	90°	a^3
BCC	$\frac{1}{2}(-1,1,1), \frac{1}{2}(1,-1,1), \frac{1}{2}(1,1,-1)$	$0.866a$	109.47°	$a^3/2$
FCC	$\frac{1}{2}(0,1,1), \frac{1}{2}(1,0,1), \frac{1}{2}(1,1,0)$	$0.707a$	60°	$a^3/4$

What I see in the cube:

- FCC: all 8 corners of the primitive box are atoms I already know: one cube corner (0,0,0), six face centers, and the opposite corner (1,1,1). The long diagonal of the box runs along [111].
- BCC: starting from a corner, three box corners land on cube corners, one on the body center, and three outside the cube, on body centers of neighboring cubes.
- Volume check: $a^3/2 = a^3 \div 2$ atoms for BCC; $a^3/4 = a^3 \div 4$ atoms for FCC.

In Materials Studio I convert with Build > Symmetry > Primitive Cell (and back with Conventional Cell). I confirm the result in the lattice parameters: FCC gives $a = b = c = 0.707a$ with 60° angles, BCC gives $0.866a$ with 109.47° angles.

2D cells of a slice

The same idea works one dimension lower: a 2D cell rebuilds one atomic slice.

1. I pick one atom in the slice.
2. I draw vectors to two nearest neighbors that do not lie on one line.
3. I complete the parallelogram.
4. I check that it holds exactly 1 atom ($4 \text{ corners} \times 1/4$).

Slice	What the cube shows	2D primitive cell
FCC (001)	Square $a \times a$ with a center atom, 2 atoms	Square of side $a/\sqrt{2}$, turned 45° , 1 atom
FCC (110)	Rectangle $a/\sqrt{2} \times a$ with edge-midpoint atoms, 2 atoms	Rectangle $a/\sqrt{2} \times a$, 1 atom
FCC (111)	Triangle of side $a/\sqrt{2}$: a cube cut, not a cell	Rhombus, side $a/\sqrt{2}$, $60^\circ/120^\circ$, 1 atom

For (111), people sometimes use a 2-atom centered rectangle ($a/\sqrt{2} \times a\sqrt{3}/2$) to get 90° axes. That rectangle is a conventional 2D cell, not a primitive one.

Rules I keep

1. A conventional cell shows symmetry; a primitive cell is the smallest working box with 1 lattice point.
2. I build either one from one atom and its nearest-neighbor vectors: 2 in 2D, 3 in 3D.

3. Area or volume of the primitive cell = conventional area or volume $\div$ atoms in the conventional cell.
4. A triangle or hexagon cut out by cube walls never counts as a cell. A cell is always a parallelogram (2D) or a parallelepiped (3D).

9. Quick reference card

- Coordinates: I put the origin at a corner, use units of a , and let the cube run from 0 to 1.
- Balls I draw: SC 8, BCC 9, FCC 14. Atoms the cell owns: 1, 2, 4.
- Plane (hkl) : cuts the axes at $1/h$, $1/k$, $1/l$. Equation $h \cdot x + k \cdot y + l \cdot z = 1$.
- (hkl) names a plane; $[hkl]$ names a direction. In cubic cells, $[hkl]$ stands perpendicular to (hkl) : $(111) \perp [111]$.
- Slice number of an atom: $s = hx + ky + lz$. Same s means same slice.
- Number of slices in a cube: I count the different values of s .
- Spacing: $d = (\text{smallest gap in } s) \times a / \sqrt{h^2 + k^2 + l^2}$.
- To see (hkl) face-on, I look along $[hkl]$ (cubic only).
- Picture A shows one slice with true distances. Picture B stacks all slices, with overlaps.
- FCC: (001) has 3 slices, $d = a/2$; (110) has 5 slices, $d = 0.354a$; (111) has 4 slices, $d = 0.577a$.
- FCC (111) projection: a hexagon with 13 spots for 14 balls; a star of two triangles plus a doubled center.
- 2D cell of FCC (111) : a rhombus, side $a/\sqrt{2}$, $60^\circ/120^\circ$, 1 atom. The triangle is a cube cut, not a cell.
- Primitive cells: SC = cube; BCC vectors $0.866a$ at 109.47° , volume $a^3/2$; FCC vectors $0.707a$ at 60° , volume $a^3/4$.